

Hongcheng Jiang

Applied AI Engineer | LLM Agents & Evaluation

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Kansas City, MO | Open to relocation / remote | U.S. Permanent Resident (no sponsorship required)

SUMMARY

Applied AI engineer with a Ph.D. in Electrical and Computer Engineering; currently a UMKC postdoctoral researcher. Builds tool-using agents from model integration to controlled execution, failure recovery, and evaluation, with research in medical imaging and clinical NLP.

TECHNICAL SKILLS

Agents: Tool Calling, MCP, Structured Outputs, Bounded Repair, Evaluation, Execution Tracing

Engineering: Python, SQL, FastAPI, SQLite, Async Workers, Docker, Linux, Git

ML: PyTorch, Hugging Face, vLLM, Ollama, Computer Vision, Diffusion Models, Vision Transformers

PROFESSIONAL EXPERIENCE

University of Missouri-Kansas City

Jun 2026 - Present

Research Associate (Postdoctoral Researcher)

Kansas City, MO

- **RadMeasure — Medical Measurement Agent:** Constrained LLM-proposed actions to registered protocols and tools; implemented geometry verification and a **KEEP/REPAIR/STOP controller** with repair budgets, independent-proposal checks, human review, and execution traces.
- **RadMeasure repair evaluation:** Across **528 runs on 176 archived cases**, intervened in **106 runs (20.1%)** with **0.69° mean error reduction among intervened runs**; overall MAE fell **2.678° to 2.540°**.
- **GeoVIRA (under review at WACV 2027):** Developed geometry-conditioned access to frozen visual features, reducing geometry-only MAE by **13.5% for hallux valgus**, **15.1% for intermetatarsal**, and **25.1% for spinal Cobb angles**; tested visual contributions with matched-versus-shuffled controls.

NextTier – IT Solutions Consultancy

Jun 2025 - May 2026

Machine Learning Engineer

Sacramento, CA (Remote)

- Built **ContractSQL**, a **Qwen3-14B natural-language SQL agent** for registered SQLite data, with browser and FastAPI/MCP interfaces for revenue totals, order lists, grouped summaries, and result review.
- Engineered schema-aware **generate–execute–repair** loops with read-only authorization, query/output limits, and up to three attempts on a consistent data snapshot; routed general answers to human review.
- Built durable asynchronous execution with idempotent jobs, renewable worker leases, retries, and **stale-worker write protection**; validated recovery using worker-kill, restart, duplicate-request, and timeout tests.
- Selected a configuration achieving **142/156 correct runs (91.0%) versus 102/156** for the single-pass Coder baseline on **26 development questions × 2 databases × 3 trials**; used asynchronous delivery for its **74.8 s p95** latency.

University of Missouri-Kansas City

Jan 2020 - May 2025

Graduate Research Assistant

Kansas City, MO

- Developed and trained transformer- and diffusion-based models for **thermal super-resolution**, **hyperspectral pansharpening**, and **NIR-to-RGB translation**; evaluated image quality and spectral fidelity.
- Co-authored a fully connected **LSTM-CRF for clinical named-entity recognition**, achieving **84.15% F1 on i2b2/VA** clinical text.

EDUCATION

University of Missouri-Kansas City

Ph.D., Electrical & Computer Engineering, GPA: 4.0/4.0

May 2025

M.S., Computer Science, GPA: 3.8/4.0

Dec 2019

PUBLICATIONS

8 peer-reviewed publications; 6 first-author imaging papers. Full list: Google Scholar.